

Lembar Pengesahan dan Identitas	Tanggal Pengajuan: 8/9/2025
Identitas Proposal	
Tahun/Periode/Skema	2025/2/KOLABORASI LUAR NEGERI
Judul	IMPLEMENTATION OF IOT-BASED FRUIT FLY TRAP DEVICES AT SALIKNETA FARM, PHILIPPINES
SDG	(2) Tanpa Kelaparan
Bidang Fokus	Pangan
TKT	6
Kelompok Keahlian	APPLIED INFORMATION TECHNOLOGY & MULTIMEDIA (KK ITEM)
Usulan Anggaran	Rp50,000,000
Koordinator Tim	
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NIP/NIDN	14870095/0423128702
Jabatan Fungsional	LEKTOR KEPALA
Fakultas/Prodi	FAKULTAS ILMU TERAPAN/PROGRAM STUDI D3 SISTEM INFORMASI (FIT)
Telepon	081283016000
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Identitas Mitra Sasaran	
Nama	Salikneta Farm
Alamat	UPPER CIUDAD REAL, SAN JOSE DEL MONTE, BULACAN, PHILIPPINES, None, None, None, None
Identitas Mitra Kolaborator	
Nama	De La Salle Araneta University and Maranatha Christian University
Alamat	De La Salle Araneta University Salvador Araneta Campus, 303 Victoneta Ave, Malabon, 1475 Metro Manila, Filipina and Maranatha Christian University Jl. Prof. drg. Surya Sumantri, M.P.H. No. 65, Bandung, 40164, Jawa Barat, Indonesia
Kontribusi Dana In Cash/In Kind	Rp32,000,000/ Rp0
Identitas Tim Pengusul	
Fakultas Tim	FAKULTAS ILMU TERAPAN
Ketua Tim	Dr. BAYU RIMA ADITYA, S.T., M.T.
Anggota Tim	RD. ROHMAT SAEDUDIN, S.T., M.T., Ph.D Dr. HELMY WIDYANTARA, S.Kom., M.Eng.

Koordinator Tim,

Ketua Kelompok Keahlian,



Dr. BAYU RIMA ADITYA, S.T., M.T.
NIP: 14870095



Dr. SUGONDO HADIYOSO, S.T., M.T.
NIP: 13870076

FORM REKOMENDASI KETUA KELOMPOK KEAHLIAN

Identitas Kegiatan	
Judul Kegiatan	IMPLEMENTATION OF IOT-BASED FRUIT FLY TRAP DEVICES AT SALIKNETA FARM, PHILIPPINES
Ketua Tim	Dr. BAYU RIMA ADITYA, S.T., M.T.
Anggota Tim	RD. ROHMAT SAEDUDIN, S.T., M.T., Ph.D Dr. HELMY WIDYANTARA, S.Kom., M.Eng.
Mitra Sasaran	Salikneta Farm
Kesesuaian Biaya	☐ Tidak sesuai ☒ Sesuai
Kesesuaian Luaran	☐ Tidak sesuai ☒ Sesuai
Kesesuaian Roadmap KK	☐ Tidak sesuai ☒ Sesuai
Rekomendasi Kegiatan	☐ Tidak direkomendasikan ☒ Direkomendasikan
Catatan Ketua KK	Roadmap sesuai dengan keilmuan AITM yaitu aplikasi sistem informasi dan IoT

Bandung, 27/8/2025



Dr. SUGONDO HADIYOSO, S.T., M.T

Abstract (Maksimal 500 kata)

Salikneta Farm, managed by the Agrivet Agricultural Science Institute of De La Salle Araneta University in Bulacan, serves as a practical training site for students and a model for the application of sustainable agricultural technology. One critical issue is fruit fly infestation, which significantly reduces crop productivity and quality. An innovative solution combines modern agricultural practices with technology. Through the implementation of IoT-based smart traps and automated monitoring systems, it is now possible to collect real-time data, analyze pest activity, and apply timely control measures. This approach not only addresses damage caused by fruit flies but also supports long-term sustainable goals.

1. Introduction

1.1. Background

Salikneta Farm, managed by the Agrivet Sciences Institute of De La Salle Araneta University in Bulacan, Philippines, is an integrated agricultural land that serves as a center for education, research, and agricultural technology development (Agrivet Sciences Institute, n.d.). The current conditions in the field show that agricultural activities at Salikneta Farm serve as an important means of practical training for students as well as a model for the application of sustainable agricultural technology, as shown in Figure 1. The farm, located in a 63-hectare area in San Jose Del Monte, Ciudad Real, Bulacan, will later be known as Salikneta, short for Saliksik sa Araneta [Research in Araneta]. However, one of the significant problems in the farm's horticultural area is the fruit fly infestation, which has the potential to reduce the productivity and quality of the harvest.



Figure 1. Salikneta Farm De La Salle Araneta University (DLSU, n.d.)

The importance of this problem lies in the significant impact that fruit fly can have on the quality and quantity of crops, as illustrated in Figure 2. In Indonesia, fruit flies from the genus *Bactrocera* (Diptera: Tephritidae) are dominant pests that attack over 150 species of fruits and vegetables, with damage rates ranging from 13.5% to 70%, and around 30% specifically for mangoes (Ariningsih et al., 2022). The importance of this problem lies in the significant impact that fruit flies have on the quality and quantity of crops. Fruit fly attacks cause fruit to rot early on, which lowers its economic value and threatens the sustainability of farming (Bausin et al., 2024; Molina-Rotger et al., 2023).



Figure 2. Mango fruit showing fruit fly damage (Syngenta Philippines, n.d.).

Another challenge is the limited use of digital technology, specifically the Internet of Things (IoT), in supporting modern and sustainable pest monitoring and control processes (Pérez Aparicio et al., 2021; Ramalingam et al., 2020; Lello et al., 2023). To this day, there is no effective and integrated pest control effort at Salikneta Farm to solve the fruit fly problem. The lack of monitoring technology and IoT-based traps means that pest attacks cannot be systematically controlled (Ahmed et al., 2024). Additionally, the limited knowledge of farmers and management related to the use of IoT agriculture further highlights the urgency of introducing applicable technological interventions (Farooq et al., 2019).

Based on these conditions, this community service program proposes the implementation of IoT-based fruit fly traps as an innovative and practical solution. Therefore, this activity not only provides a means of mitigating fruit fly problems, but also serves as a medium for transferring knowledge related to IoT technology in agriculture. The implementation of these traps is hoped to support Salikneta Farm as a center for modern agricultural education and provide real benefits to the surrounding community in increasing their awareness and skills in encountering fruit fly pest challenges in an adaptive way.

1.2. Potential Development of Target Community

A major challenge for the agricultural community at Salikneta Farm, Bulacan, Philippines, is the prevention and control of pests on the farm, particularly fruit flies, which are a major enemy for farmers. This is because Salikneta Farm comprises fruit orchards, vegetable production areas, and livestock. Lack of fruit fly prevention and control practices by farmers can lead to economic losses due to crop damage and reduced quality. Larvae hatching from eggs inside the fruit feed on the flesh, causing rot and loss. An accurate understanding of soil nutrition and environmental conditions is a major obstacle to supporting local farmers. Therefore, sustainable solutions are urgently needed.

Developing an IoT-based fruit fly trap could be an effective solution. A more targeted approach to implementing the IoT-based fruit fly trap will provide more measurable and accurate capabilities in preventing and controlling fruit flies that attack various crops on Salikneta Farm.

Through this community service program, we are presenting a solution in the form of implementing an IoT-based fruit fly trap. This system will help the farming community at Salikneta Farm, Bulacan, Philippines, provide more effective support to farmers in the prevention and control of fruit fly pests.

The use of appropriate technology not only provides access to information but also empowers farmers to manage and utilize this technology independently. Thus, this program



not only offers solutions to agricultural problems but also represents a step towards improving the quality of services and the effectiveness of the farming community at Salikneta Farm, Bulacan, Philippines.

2. Proposed Solution

Proposed Community Service Solution

The implementation of Internet of Things (IoT)-based fruit fly traps at Salikneta Farm is to help increase productivity and maintain crop quality through more effective pest control.

The device used is a smart trap with sensors, infrared cameras, and wireless communication modules to monitor fruit fly populations in real time. The data collected not only shows the number of insects, but can also be processed to figure out the daily activity patterns of fruit flies so that control measures can be applied at the right time.

The device design uses Orange Pi 5 as a local processor connected to a NoIR camera and infrared LED lights, enabling it to automatically record images day and night, as illustrated in Figure 3. The captured data is stored in local memory and then periodically transmitted via Wi-Fi to a cloud-based server. This server is completed with an analytics system to process the captured data and display pest population visualizations through a dashboard accessible to Salikneta Farm managers.

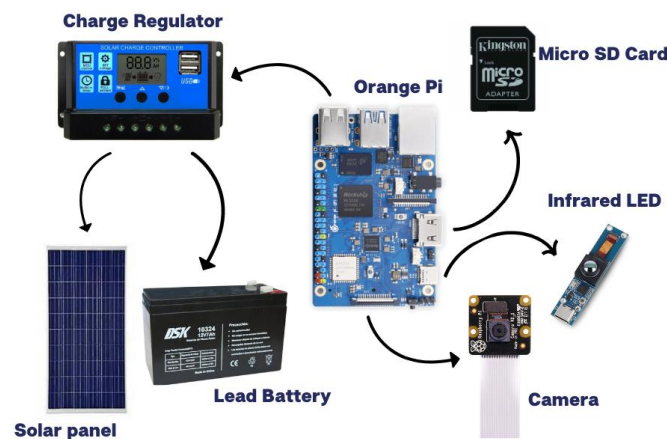


Figure 3. IoT-Based Smart Trap for Fruit Flies

The smart trap is made from a modified polypropylene plastic container that is weatherproof. The inside of the box is equipped with sticky cards to trap insects, with small ventilation holes to allow rainwater to escape. The camera is positioned to clearly capture the entire surface of the sticky card. The device's power source comes from a solar panel connected to a rechargeable battery and a current regulator, enabling the trap to operate long-term without relying on conventional electricity.

The trap works by using pheromones placed inside the box to attract fruit flies. The insects that enter then stick to the sticky cards that have been placed on the floor of the trap. The camera periodically takes pictures, both day and night, and the images are analyzed to count the number of fruit flies caught. The capture data is stored, processed, and sent to a server, enabling remote monitoring of pest populations via a dashboard. The trap operates automatically, from attracting pests, capturing them, recording the data, to sending the data.

The dashboard connected to the smart trap system serves as an information center for Salikneta Farm. Through this dashboard, fruit fly capture data is displayed in an easy-to-understand visual format, such as daily, weekly, and seasonal population trend graphs. In addition to the number of captures, the dashboard can also show the peak activity time of fruit flies based on image analysis, thereby helping to determine the most suitable period to conduct control measures.

With weather-resistant design, solar power, and cloud-based analytical systems, this smart trap is supported to provide more precise pest monitoring, reduce reliance on chemical



insecticides, and support the implementation of sustainable agricultural practices at Salikneta Farm.

3. Methodology (Maksimal 2000 kata)

3.1. Methods and Steps

Community service activities are implemented in several stages, including:

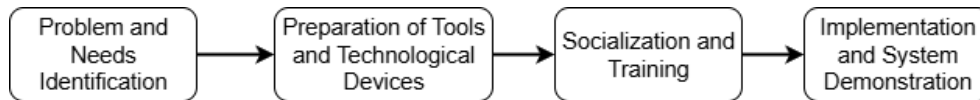


Figure 4. Methods and Steps

1. **Problem and Needs Identification**
The initial stage focused on gathering information related to the problems and needs encountered by the partners. The identification process was conducted through discussions with Salikneta Farm. From the results of this identification, the need for technological solutions was identified
2. **Preparation of Tools and Technological Devices**
At this stage, the team will prepare IoT-based fruit fly traps. The devices will be assembled and configured to ensure proper integration between hardware and software components, as well as compatibility with environmental conditions on the farm
3. **Socialization and Training**
Socialization activities were conducted to increase awareness at Salikneta farm about the importance of fruit fly traps using technology. Next, training will be provided on how to use the equipment. This stage aims to ensure that partners are capable of operating and utilizing the technology independently
4. **Implementation and System Demonstration**
The final stage is hands-on implementation at the Salikneta Farm by deploying fruit fly traps. Hands-on demonstrations will also be provided to partners, along with joint evaluations to assess the success of the implementation and identify opportunities for future improvements.

3.2. Target Community Participation

The participation of target community partners in the IoT-Based Fruit Fly Trap Development Project in the Slikneta Farming Community, Bulacan, Philippines, will cover several aspects critical to the program's success:

1. During the technology development phase, partners, including the farming community at Salikneta Farm, Bulacan, Philippines, will be involved in identifying system requirements and specifications. They will provide insight into the challenges and needs faced in controlling pests on farms
2. During implementation, farmers and the farming community will act as direct users of the IoT-Based Fruit Fly Trap. Their participation in the use of the device will be key to the program's success
3. They will help identify farmers' needs and preferences regarding the technology being developed
4. They will play a critical role in validating the usability and efficacy of the developed device
5. In addition, the farming community at Salikneta Farm, Bulacan, Philippines, will serve as extension agents and key advocates in promoting the use of the IoT-Based Fruit Fly Trap to farmers and the local community. They will disseminate information about the benefits and how to use this tool, as well as assist in user training and mentoring.

Meanwhile, the contributions of the collaborative partners, namely Araneta De La Salle University and Maranatha Christian University, are as follows:

1. Contributing to the design development of IoT technology devices in the target areas
2. Providing technical support during the system implementation and maintenance phase

3. Providing some resources and financial support for the implementation of this community service activity. Financial support from these collaborative partners will help ensure the smooth and sustainable implementation of the IoT-Based Fruit Fly Trap tool in the farming community at Salikneta Farm, Bulacan, Philippines.

3.3. Sustainability Opportunity of the Program

To ensure the sustainability of the program, the following steps will be taken:

- Utilization for other relevant pest types in the Salikneta Farm agricultural area
- Development of an early warning system for farmers
- Development of a dashboard and AI features.

Based on the 2024-2028 Applied IT & Multimedia Program priorities, the proposed solution for developing an IoT-based Fruit Fly Pest Trap is related to the 2025 priority, namely as a digital transformation model for smart agriculture.

Peta Jalan Penelitian sub KK Sistem

Informasi



Figure 5. KK AITM Roadmap

4. Output Target

The output target of this community service is summarized in:

Output	Type	Indicator
Compulsory Output	1. Academic Publication (published in national or international journal) or Intellectual Property Right; 2. News Published in external mass media; 3. Video	1. Academic Publication 2. News Published in external mass media 3. Video
Additional Output	1. Modul	Module in soft file



5. Budget Plan And Schedule			
5.1. Budget Plan			
5.1.1. Consumable Materials (A)			
No.	Component	Detail	Total
1	Meeting Consumption	6 Persons × (30,000 Lunch + 15,000 Snack)	270,000
2	Meeting Consumption	6 Persons × (30,000 Lunch + 15,000 Snack)	270,000
3	Meeting Consumption	6 Persons × (30,000 Lunch + 15,000 Snack)	270,000
4	Meeting Consumption	6 Persons × (30,000 Lunch + 15,000 Snack)	270,000
5	Souvenirs	4 Packs x 150.000	600,000
6	Community Service Banner	1 Banner	135,000
7	Alat Perangkap Hama IoT: Microcontroller, Orange Pi, Kamera, Lampu LED, Panel Surya, Micro SD, Charge, Tiang, Komunikasi Data, dll	Paket	12,255,000
8	Pendukung Instalasi: Kabel, Semen, Solasi, Ember, dll	Paket	2,000,000
Total (A)			16,070,000
5.1.2. Transportation (B)			
No.	Component	Detail	Total
1	Travel Tel-U - CKG	4 Orang	774,000
2	Travel CKG -Tel-U	4 Orang	794,000
3	Pesawat Terbang (Kelas Ekonomi)	4 Orang	16,000,000
4	Tarif Hotel	4 Kamar 2 Malam	3,600,000
5	Uang Harian Jabatan Akademik Dosen	3 LK negara C 5 Hari	7,500,000
6	Uang Harian Mahasiswa	5 Hari	1,000,000
7	Grab Hotel - Salikneta Farm	2x Perjalanan	116,000
8	Grab Salikneta Farm - Hotel	2x Perjalanan	116,000
9	Grab Bandara - Hotel	1x Perjalanan	170,000
10	Grab Hotel - Bandara	1x Perjalanan	170,000
Total (B)			30,240,000



5.1.3. Administration & Dissemination (C)			
No.	Component	Detail	Total
1	Biaya Jurnal Abdimas	500.000/Jurnal	500,000
2	Biaya Tenaga Penunjang Peneliti - IoT	110.000/OH x 19	2,090,000
3	Biaya Tenaga Penunjang Peneliti - Desain Produk	110.000/OH x 10	1,100,000
Total (C)			3,690,000
Total Proposed Budget (A+B+C)			50,000,000

5.2. Schedule												
The following is the activity schedule for the execution of the program:												
Activities	Months											
	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26	Jun-26	Jul-26	Aug-26
Problem & Needs Identification												
Preparation of Tools & Devices												
Socialization & Training												
Demonstration & Implementation												
Final Evaluation												

6. References
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- Ahmed, S., Marwat, S. N. K., Brahim, G. B., Khan, W. U., Khan, S., Al-Fuqaha, A., & Koziel, S. (2024). IoT based intelligent pest management system for precision agriculture. *Scientific Reports*, 14(1), 31917.
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7. Attachment

7.1. Surat Persetujuan/Pernyataan Masyarakat Sasar

De La Salle Araneta University
Agrivet Sciences Institute
Salikneta Farm, Upper Ciudad Real
San Jose del Monte, Bulacan, Philippines

Letter of Support for Collaboration

The undersigned below:

Name: **Jennifer DC. Bandoja**

States that she is willing to be a partner for Telkom University's Internal Community Service Grant Scheme (International). She will join the community service project entitled "*Implementation of IoT-Based Fruit Fly Trap Devices at Salikneta Farm, Philippines*" with Bayu Rima Aditya and team from Telkom University and Oscar Karnalim and team from Maranatha Christian University.

 30 August 2025
Jennifer DC. Bandoja
PRINTED NAME AND SIGNATURE



7.2. Letter of Statement

LETTER OF STATEMENT

According to the Agreement between **Telkom University (TEL-U), De La Salle Araneta University, and Maranatha Christian University**, all universities agree to provide contribution for this International Community Service Program.

Both universities have agreed with the following statements:

1. The funding amount for TEL-U, De La Salle Araneta University, and Maranatha Christian University international community service program 2025 is 000 USD with details as follows:

No.	Institution	Project Title	Project Leader	Amount of Contribution
1	Telkom University	Implementation of IoT-Based Fruit Fly Trap Devices at Salikneta Farm, Philippines	Dr. Bayu Rima Aditya, S.T., M.T.	IDR 50.000.000 or 3000 USD (in cash)
2	De La Salle Araneta University		Mr. Elmer R. Montebon	IDR 16.000.000 or 1000 USD (in cash)
3	Maranatha Christian University		Oscar Karnalim, S.T., M.T., Ph.D.	IDR 16.000.000 or 1000 USD (in cash)

2. Telkom University - (collaborator's name) international community service program 2025-2 will be officially started on (September), 2025.

Telkom University

De La Salle Araneta University

Maranatha Christian University

Dr. Kemas Muslim L.
Director
Research and Community Service

Mr. Elmer Montebon, M.Sc., R.Agr.
Program Chair, Agriculture
College of Veterinary Medicine
and Agricultural Sciences

Oscar Karnalim, S.T., M.T., Ph.D., SMIEEE, CSCU.
Dean, Faculty of Smart
Technology and Engineering,
Maranatha Christian University

7.3. Letter of Agreement



**LETTER OF AGREEMENT / STATEMENT OF COLLABORATOR PARTNER
COMMUNITY SERVICE
TELKOM UNIVERSITY**

I am the undersigned:

Name : Elmer R. Montebon
Position : Lecturer/Assistant Professor
Agencies : De La Salle Araneta University

Declare our confirmation as a partner of Community Service activities:

Title : Implementation of IoT-Based Fruit Fly Trap Devices at Salikneta Farm, Philippines

With the following contributions:

Contribution Type	Nature of Contribution	Amount (Rp)
a. Funding	<i>In cash</i>	Rp. 16.000.000 (1000 USD)

Thus we make this letter to be used as it should.

Malabon City, August 27, 2025



Mr. Elmer R. Montebon
Program Chair of Agriculture
College of Veterinary Medicine and Agricultural Sciences



UNIVERSITAS
KRISTEN
MARANATHA

Fakultas Teknologi dan Rekayasa Cerdas

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**LETTER OF AGREEMENT / STATEMENT OF COLLABORATOR PARTNER
COMMUNITY SERVICE
TELKOM UNIVERSITY**

I am the undersigned:

Name : Oscar Karnalim, S.T., M.T., Ph.D.
Position : Dean of The Smart Technology and Engineering Faculty
Agencies : Maranatha Christian University

Declare our confirmation as a partner of Community Service activities:

Title : Implementation of IoT-Based Fruit Fly Trap Devices at Salikneta
Farm, Philippines

With the following contributions:

Contribution Type	Nature of Contribution	Amount (Rp)
a. Funding	<i>In cash</i>	Rp. 16.000.000 (1000 USD)

Thus we make this letter to be used as it should.

Bandung, 26 August 2025



Oscar Karnalim, S.T., M.T., Ph.D.
(Dean of The Smart Technology and Engineering Faculty)